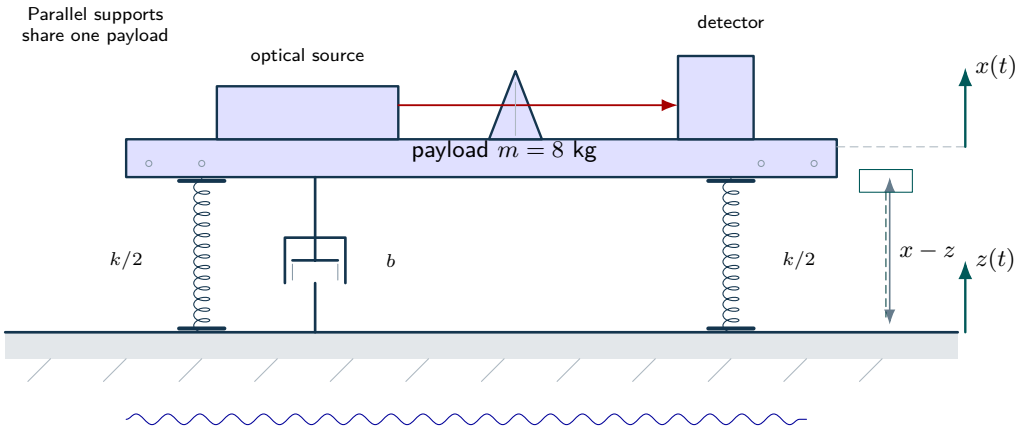

Moving floor; displacement measured from static equilibrium

Measured coordinates

x, z: absolute displacements [m].
$x - z$: spring travel [m].
$\dot{x} - \dot{z}$: damping velocity.

Force balance

$$m\ddot{x} + b(\dot{x} - \dot{z}) + k(x - z) = 0.$$

Two $k/2$ supports give total k.

Operating boundary

Coordinates are deviations about static sag. The blue floor trace indicates motion; it is not a measured waveform.